The due date for submitting this assignment has passed.
Due on 2026-03-04, 23:59 IST.

You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-03-04, 08:54 IST

In this activity, we will go through a problem based on the Page Rank Algorithm, introduced by Larry Page and Sergey Brin, in order to rank web pages in search engine results. The fundamental idea behind the page rank algorithm is that the importance of a webpage, which will be called its page rank, is derived from the number and quality of links pointing to it. For simplicity, we consider a small collection of four web pages and describe the procedure used to determine the rank of each page.

The following data is known:
Page 1 has links to Page 2 and Page 4.
Page 2 has links to Page 1, Page 3 and Page 4.
Page 3 has links to Page 1 and Page 2.
Page 4 has links to Page 1 and Page 2.

1) **Step I. Constructing a directed graph using the linking information.** ***1 point***

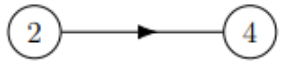
First, we obtain a directed graph whose vertices are the web pages in the collection. There is a directed edge from a Page i to Page j iff there is a link from Page i to Page j. For the given information, choose all edges which belong to the graph as constructed above

- [x] 

- [ ] 

- [ ] 

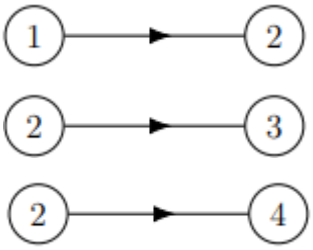
- [x] 

- [x] 

Yes, the answer is correct.

Score: 1

Accepted Answers:



2) **Step II. Obtaining the transition matrix.** ***1 point***

For each Page i, where $i \in \{1, 2, 3, 4\}$, we associate a real number x_i to it, which is its page rank mentioned above. Recall that the page rank of a page is an indicator of the importance of the page. Thus, we make the following natural assumptions on the page ranks.

· Every page distributes its own importance equally to all its links. In other words, if there is a link from page i to page j, and there are k links in page i, then page i contributes $\frac{1}{k}x_i$ page rank to the page rank of j.

· The page rank of every page is equal to the sum of the contributions of the page ranks of all pages that link to it, where the contributions are computed using the previous rule.

For example, for the given system, page 1 gets its page rank from page 2, page 3 and page 4. Analysing the number of outgoing links from each page, we get

$$x_1 = \frac{1}{3}x_2 + \frac{1}{2}x_3 + \frac{1}{2}x_4.$$

Doing the same for all pages, we obtain a system of equations which can be written as

$$P\tilde{x} = \tilde{x},$$

where $\tilde{x} = (x_1, x_2, x_3, x_4)^T$. P is called the transition matrix. It can also be checked that the $(i, j)^{th}$ entry of P is the probability that a user lands at page i using a link from page j. Which of the following statements are true for the transition matrix P obtained using the given data?

- [ ] The first column of P is $(0, \frac{1}{2}, \frac{1}{2}, 0)^T$.
- [x] The second column of P is $(\frac{1}{3}, 0, \frac{1}{3}, \frac{1}{3})^T$.
- [x] The third column of P is $(\frac{1}{2}, \frac{1}{2}, 0, 0)^T$.
- [ ] The fourth column of P is $(\frac{1}{2}, 0, \frac{1}{2}, 0)^T$.

☑ The sum of the entries of each column of P is 1.

Yes, the answer is correct.

Score: 1

Accepted Answers:

The second column of P is $(\frac{1}{3}, 0, \frac{1}{3}, \frac{1}{3})^T$.
The third column of P is $(\frac{1}{2}, \frac{1}{2}, 0, 0)^T$.
The sum of the entries of each column of P is 1.

3) **Step III. Finding the solution set.**

The system of equations mentioned above can be rearranged to obtain a homogeneous system of linear equations. If the set of solutions to the system of equations is given by

$$\{(8t, ct, 3t, 7t) \mid t \in \mathrm{R}\},$$

then find the value of c.
9

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 9

1 point

4) **Step IV. Finding the page ranks.**

There are infinitely many solutions to the system of equations that we obtain from the given information. However, there is a unique solution if we also impose the condition that the sum of the entries of the solution should be 1 (This is also related to the probability of landing on any page). The page rank of a page i is defined as the ith entry in the solution (x_1, x_2, x_3, x_4) such that $\sum_{i=1}^{4} x_i = 1$. Find the page rank of the page with the minimum rank. (Enter your answer rounded to two decimal places.)
0.11

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 0.11

1 point